

C and C++

The oddly named C and C++ are two of the most popular programming languages in existence. They have been used to create a huge amount of software we use today.

The C programming language

American computer scientist Dennis Ritchie (1941–2011), a programmer at Bell Labs in the USA, released C in 1978. He developed the language while working on the Unix operating system. Unix was coded in assembly language, which tied it to a particular type of computer. This meant that the number of customers willing to buy it were limited. Ritchie created C so that a new version of UNIX could be made that could run on any machine.

```
Claire's-MacBook-Air:C claire$ clang -wall hello -o hello
Claire's-MacBook-Air:C claire$ ./hello
Hello, World!
```

Command line

SEE ALSO

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◀ 110–111 Translation

◀ 118–119 What do programming languages do?

◀ 120–121 Types of programming language

▽ IDE

An integrated development environment (IDE) lets programmers write, compile, and run code using a single program with a graphical interface. IDEs make writing large software systems a more manageable process.

```
#include <stdio.h>

int main()
{
    printf("Hello, World! \n");
    return 0;
}
```

IDE

How does it work?

C is an imperative programming language and doesn't allow object-oriented or functional styles of programming. C's syntax, using curly braces { } to enclose blocks of code, has influenced many other languages. It's a high-level language that doesn't abstract away from the internal structure of the computer. This means programmers can directly access areas in a computer's memory.



Computer operating system



Robotics



NASA's core flight system



Arduino microprocessor

△ What is it used for?

C's combination of high- and low-level features make it popular for writing operating systems, particularly for the most essential parts. Given its flexibility, it is used in a huge variety of applications.

LINGO

Behind the names

The language Ritchie originally tried to re-implement Unix in was called B, short for BCPL (Basic Computer Programming Language). C was simply the next letter in the alphabet. Putting “++” after a variable in C tells the computer to add one to it, so $1++ = 2$. The name C++ reflects the fact that the new language is C, but with additions.

This code sets a variable called “age” to 20, and then by asking for “age++” the number 21 is displayed.

```
int age = 20;
printf("Age is: %d", age);
age++;
printf("Age is now: %d", age);
```